

Alpilla 600 MW Combined Cycle Power Plant Materials Selection and Corrosion Control Report

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Originator	Istanbul EPC Mühendislik ve Taahhüt A.Ş.
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1. Purpose and scope

1.1 This report defines the environment classification of the site and the selection of materials and corrosion protection measures for the equipment, piping and structures of the 1 x 1 combined cycle power plant, for a design life of 30 years, exposed to a marine C5 atmosphere and to seawater from the Marmara coast.

1.2 The report is the basis of the material entries of the equipment register and of the piping material classes (ALP-EPC-00000-PI-SPC-0001), the painting specification (ALP-EPC-00000-ME-SPC-0002) and the cathodic protection design (ALP-EPC-00000-EL-CAL-0005). Materials in vendor proposals shall be reviewed against this report.

1.3 Material selection is preliminary until the vendors' data and the seawater analysis of the detailed design are available; where the report states the choice open (e.g. GRP or HDPE for hypochlorite storage), the selection shall be recorded in the equipment register before IFC issue of the datasheet.

2. Governing requirements and decisions

ER-01.03: All equipment and structures shall be designed for an economic life of 30 years, 8,000 equivalent operating hours per year and up to 250 starts per year (of which up to 50 cold, 100 warm and 100 hot starts), including daily cycling and load following.

ER-02.04: Seawater quality, temperature and marine conditions per ALP-OWN-MAR-001 (two-layer sea, pycnocline at 20 m to 25 m). Design range at the intake depth 6.9 degC to 27.0 degC, chloride up to 14,500 mg/l, TSS up to 140 mg/l in storms, occasional H₂S in the lower layer, mussel settlement, jellyfish blooms up to 20 kg/m³ and mucilage events.

ER-02.08: Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

ER-08.03: 2x50 % vertical CW pumps with a common spare pump assembly stored on Site; materials suitable for seawater (duplex).

ER-08.04: Electro-chlorination with intermittent/shock dosing; residual chlorine at outfall ≤ 0.1 mg/l.

ER-15.04: Material certificates per EN 10204 3.1 (3.2 for pressure parts above 50 bar) and positive material identification of alloy piping.

DEC-EPCE-0015 Cycle chemistry, seawater chlorination and water treatment concept: Cycle chemistry AVT(O) per VGB-S-010 / IAPWS TGD (ammonia only, no oxygen scavenger in operation); the oxygen scavenger dosing is kept for wet lay-up only; phosphate / caustic trim dosing of the HP drum for upsets. Condensate polishing 2 x 100 % mixed-bed vessels, resin exchange by a service contractor (no on-site regeneration). Seawater electro-chlorination: continuous low-level dosing 0.5 mg/l at the intake with pulse dosing to 2 mg/l for 3 x 30 min per day, residual oxidant (TRO) measured at the condenser outlet and at the outfall seal pit; dosing controlled on the seal-pit TRO with set-point 0.08 mg/l and a high alarm at 0.1 mg/l, the compliance point being the end of the outfall pipe (DP-CASE-0017, TQ-041); pulse dosing is cut back when the seal-pit TRO reaches the set-point; generator capacity 60 kg/h Cl₂ (1.5 mg/l on 38,000 m³/h), average consumption about 25 kg/h; no dechlorination. SWRO 2 x 100 % with isobaric energy recovery, brine returned to the CW outfall seal pit only while at least one CW pump runs (interlock: the SWRO trips on loss of all CW pumps, the demineralised water storage 2 x 800 m³ covers the outage), dilution at least 1:200 with both SWRO trains and one CW pump, EDI reject recycled to the SWRO

feed. Potable water from SWRO permeate with calcite remineralisation and hypochlorite residual chlorination. Neutralisation of the waste water with sulphuric acid and caustic dosing in the waste water treatment package.

3. Environment classification

3.1 Atmospheric corrosivity of the site is category C5 (very high, marine) per ISO 12944-2 with durability class H (over 15 years to first major maintenance). Splash zones, areas constantly wetted by seawater spray and immersed structures are classified CX / Im2 per ISO 12944-2 for the purposes of this report.

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001
Chloride deposition maximum	420 mg/m2/day	ISO 9225, category S3	confirmed	SRC-OWNR-0001, ER-02.10
Ground flash density	2.4 flashes/km2/year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001, ER-02.12
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001, ER-02.11
PM10 maximum (event)	260 ug/m3	Saharan dust episodes	confirmed	SRC-OWNR-0001, ER-02.10
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001
Sea-salt aerosol annual mean	14 ug/m3	as NaCl	confirmed	SRC-OWNR-0001
Sea-salt aerosol maximum (event)	165 ug/m3	as NaCl, lodos storms	confirmed	SRC-OWNR-0001, ER-02.10
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001, ER-02.11
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001, ER-02.11
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels >= +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m2		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

3.2 The chloride deposition rate (maximum, DP-OWNR-0007) and the sea-salt aerosol concentrations (annual mean DP-OWNR-0006, event maximum DP-OWNR-0005) are the design basis for equipment enclosures, air intakes, insulation cladding and electrical equipment. Air intake filtration of rotating machines shall be designed for the sea-salt event value, including fine salt mist and PM10 events (DP-OWNR-0008).

3.3 Seawater used for once-through cooling shall be assumed to have chloride concentration up to the maximum of DP-OWNR-0014, temperature range per DP-OWNR-0011 to DP-OWNR-0013, suspended solids up to the storm value of DP-OWNR-0017 and a jellyfish load per DP-OWNR-0019. Salinity is low (brackish, DP-OWNR-0015); the design shall however assume full-strength chloride corrosivity for pitting and crevice corrosion, since salinity varies with the pycnocline layer.

3.4 Continuous chlorination of the seawater at 0.5 mg/l with pulse dosing to 2 mg/l, and a residual oxidant of up to 0.2 mg/l in the condenser and the CW outlet piping (dosing controlled on the outfall seal-pit value, DEC-EPCE-0015), shall be taken as the oxidant exposure of all wetted materials downstream of the dosing point; materials shall be tolerant of free chlorine at those levels without a loss of pitting resistance.

Parameter	Value	Condition	Status	Basis
Condenser cleanliness factor (design)	0.85 -	HEI	confirmed	ER-07.07
Condenser cooling water temperature rise	7 K	maximum, base load	confirmed	ER-07.04
Condenser pressure at SRC (maximum)	36 mbar(a)	base load, seawater 16.0 degC	confirmed	ER-07.08
Condenser tube velocity	1.8 to 2.2 m/s	titanium	confirmed	ER-07.07
CW intake velocity at the screens	0.15 m/s	maximum, with fish return system	confirmed	SRC-OWNR-0007
Jellyfish load design	20 kg/m3	blooms June-September	confirmed	SRC-OWNR-0002, ER-08.07
Pycnocline depth	20 to 25 m		confirmed	SRC-OWNR-0002
Residual chlorine at outfall	0.1 mg/l	maximum	confirmed	ER-08.04
Seawater boron	2.4 to 3.4 mg/l		confirmed	SRC-OWNR-0002
Seawater chloride - maximum (upper layer)	14,500 mg/l		confirmed	SRC-OWNR-0002
Seawater salinity range (upper layer)	21.8 to 26.1 PSU	lower layer 38.2 PSU below pycnocline	confirmed	SRC-OWNR-0002
Seawater temperature at intake - annual mean (measured)	15.4 degC	12 m depth, station M2	confirmed	SRC-OWNR-0002
Seawater temperature at intake - maximum	27 degC	August	confirmed	SRC-OWNR-0002
Seawater temperature at intake - minimum	6.9 degC	February	confirmed	SRC-OWNR-0002
Seawater temperature range at intake	6 to 27 degC		confirmed	ER-02.04
Seawater TSS maximum (storm)	140 mg/l		confirmed	SRC-OWNR-0002
SRC seawater temperature	16 degC	Site Reference Conditions	confirmed	ER-02.03
SWRO brine salinity excess at 50 m from the diffuser	5 %	above ambient salinity, maximum	confirmed	SRC-OWNR-0007
Thermal plume excess at 100 m mixing zone	3 K	maximum	confirmed	ER-08.05
Wave height Hs 50-year	4.4 m	Tp 8.5 s, SW	confirmed	SRC-OWNR-0002

3.5 Buried and immersed items shall be designed for the soil conditions and groundwater level of the site geotechnical report (groundwater DP-OWNR-0047); the soil resistivity and aggressiveness shall be confirmed in the geotechnical survey before the design of underground metal piping and the cathodic protection.

4. Materials by service

Item	Value / rule	Remarks
Seawater, pumps, valves, strainers (wetted)	Super duplex (UNS S32750 / S32760, PREN >= 40), 6Mo or titanium Gr 2; rubber-lined carbon steel for large casings and columns	Per ER-08.03; no 316L in stagnant chlorinated seawater [ER-08.03, ER-02.04]
Seawater, condenser tubes and cooler plates	Titanium Gr 2 tubes and plates; titanium-clad tube sheets; EPDM or FKM gaskets	Consistent with the equipment register for condenser and CCW coolers [ER-02.04]
Seawater, large piping (CW mains, intake, outfall)	GRP (vinyl ester resin) underground and above ground; rubber-lined carbon steel for fittings with flanges DN 800 and above where pressure class requires	Piping class per ALP-EPC-00000-PI-SPC-0001 [ALP-EPC-00000-PI-SPC-0001]
Seawater, small bore piping (electro-chlorination, sample, wash water)	Super duplex or GRP / PVC-U / PP where pressure and temperature allow	Avoid 316L; PVC only below 40 degC
Cycle water and steam, feedwater and condensate	Carbon steel (SA-106 Gr B / C) up to 400 degC; P22 (SA-335) for cold reheat and IP steam; P91 (SA-335) for HP and hot reheat steam; 13Cr or cast steel for condensate pump internals	Consistent with tie-in data; erosion-corrosion: chromium content >= 0.1 % in wet / two-phase lines
Condensate and demineralised water	Carbon steel with 316L / 304L in demin storage and wetted parts of demin pumps; PP / PVC-U / 316L in the demin package	AVT(O), ammonia, pH 9.2 to 9.6 per DEC-EPCE-0015 [DEC-EPCE-0015]
Ammonia (aqueous 24.5 %)	Carbon steel (API 650) for the atmospheric storage tanks 10HSB10BB001/2; 316L for pumps, valves and piping; PTFE seals; no copper, zinc, brass or bronze in wetted parts	Copper alloys attacked by ammonia (stress corrosion cracking); 304L excluded outdoors (section 5) [DEC-EPCE-0015]
Sodium hypochlorite (0.8 to 1 % on-site generated)	GRP (vinyl ester) or HDPE tanks; PVC-U / PP / PVDF piping; titanium wetted parts of pumps	No stainless steel; no carbon steel [DEC-EPCE-0015]
LDO and lubricants	Carbon steel; stainless steel filters and elements; no zinc-coated surfaces in wetted parts; no copper in injectors	Sodium and potassium contamination limits per DEC-OWNR-0001
Fuel gas	Carbon steel (LTCS where pressure reduction gives low temperature); 316L for instrument tubing	H2S and sulphur limits of the gas specification; wet gas not expected
Chemical dosing (acid, caustic, phosphate, corrosion inhibitor)	316L, PVDF, PP, PTFE-lined, HDPE by chemical; no aluminium with caustic	Selection by chemical per supplier data
Fire water	Ductile iron, carbon steel galvanised or GRP for underground; bronze or stainless internals of pumps and valves (fresh-water fire water)	Fire water is SWRO permeate, no seawater admitted (DEC-EPCE-0018) [DEC-EPCE-0018]
Potable water and SWRO permeate	GRP, HDPE, PVC-U, 316L; internally lined ductile iron for underground pipe	Materials approved for potable use

Item	Value / rule	Remarks
Stainless steel selection rule	304L only in dry indoor or freshwater service; 316L in outdoor service with insulation excluded from chloride contact; duplex or super duplex for chloride exposure at above 60 degC in the marine atmosphere	Chloride stress corrosion cracking limit for austenitic steels under insulation [REF-CONS-0092]

5. Marine atmosphere: carbon steel, coating and stainless steel

5.1 Outdoor carbon steel equipment and structures shall be coated with a C5 coating system of durability H in accordance with ALP-EPC-00000-ME-SPC-0002; hot-dip galvanising per EN ISO 1461 for structural steel, gratings and supports in accordance with ER-13.03, with an additional coating in the splash zone and where chloride deposition is above the maximum stated in DP-OWNR-0007.

5.2 Uncoated stainless steel in the outdoor marine atmosphere shall be limited to super duplex, 6Mo and 316L with surface roughness Ra of 0.5 micrometre or better for exposed surfaces; 304 / 304L is not permitted outdoors. Where stainless steel is used for small bore piping, tubing and instrument fittings outdoors, 316L with electropolishing or a coating shall be used, and the tubing shall be of the grade 6Mo or 2507 in the splash zone.

5.3 Coating versus stainless steel: for carbon steel items above 2 m² exposed surface in C5, coating (ISO 12944, system for durability H) is the default; stainless steel or duplex is selected where the maintenance access for repainting is not possible for 30 years, in seawater immersion, and for small items (fasteners, tags, instrument tubing).

5.4 Fasteners outdoors shall be A4-80 (316) as a minimum; bolts of flanges on carbon steel piping with coating may be hot-dip galvanised (EN ISO 10684) with PTFE coating on the thread; A4-80 shall not be paired with aluminium unless isolated.

5.5 Aluminium and GRP items outdoors (fan impellers, louvres, cable trays) shall be checked for chloride attack; aluminium alloy 5xxx or 6xxx with anodising and paint is accepted for non-structural parts. Cable ladders shall be hot-dip galvanised or of GRP; stainless steel 316L in the splash zone.

5.6 Equipment enclosures (electrical, instrument) shall be of stainless steel 316L, aluminium alloy or steel with a C5 coating; enclosure seals and gaskets shall be of EPDM or silicone, and internal parts shall have the C5 marine protection stated in the equipment register.

6. Galvanic corrosion and isolation

6.1 Dissimilar metals in contact in an electrolyte (seawater, condensate on surfaces) shall not be used unless the galvanic potential difference is below 0.15 V, or the anodic part is more than 10 times the area of the cathodic part, or the contact is isolated. Isolation shall be achieved with isolating flanges (gaskets, sleeves and washers of PTFE / G10), non-conductive spools of GRP at least 5 diameters long, or lined pipe.

6.2 Titanium condenser tubes and CCW plates shall be isolated from carbon steel components in contact with seawater; carbon steel tube sheets shall have titanium cladding; water box and channel rubber lining shall be continuous with no exposed carbon steel in seawater contact.

6.3 Stainless and super duplex components inside rubber-lined carbon steel bodies (discs, shafts, seats) shall be isolated with the lining; sacrificial anodes (zinc or aluminium) shall be provided in seawater boxes and pump columns where the coating or lining can be damaged, with the design per the cathodic protection design.

7. Cathodic protection

7.1 Cathodic protection of the seawater intake structure, pump house metalwork, outfall and buried carbon steel piping shall be designed in accordance with ALP-EPC-00000-EL-CAL-0005 (Cathodic Protection Design) for a design life of 30 years, with the current density and anode mass based on the seawater data of this report and on the coating breakdown factors of the coating specification.

7.2 Sacrificial anode systems shall be used for immersed structures and seawater equipment; impressed current systems shall be considered for buried piping only where the anode mass calculation shows more than 40 % saving over the design life. The interaction between the earthing system and the cathodic protection of buried steel shall be checked in the earthing design.

7.3 Electrical isolation between the cathodically protected structures and the plant earthing shall be provided by isolating joints with surge protection (polarisation cells or decoupling devices); isolation shall be tested before commissioning.

8. Corrosion allowances

Item	Value / rule	Remarks
Carbon steel, dry steam, boiler feed and condensate	1.5 mm	Erosion-corrosion checked for wet steam and two-phase lines
Carbon steel, LDO, fuel gas, compressed air, nitrogen	1.0 mm	0 mm for fuel gas piping upstream of the fiscal meter if the gas is dry
Carbon steel, cooling water (closed circuit), demin water, drains	3.0 mm	Coated or lined where the allowance is otherwise not adequate
Carbon steel, fire water and raw water	3.0 mm	Internally lined or galvanised underground piping
Carbon steel tank bottoms and shell in LDO service	1.5 mm shell; 3.0 mm bottom (internal coating and cathodic protection under bottom)	API 650 [REF-CONS-0066]
Rubber-lined or coated carbon steel in seawater	3.0 mm on the structural steel in addition to the lining	Lining is not a corrosion allowance
Stainless steel, duplex, super duplex, titanium	0 mm	No allowance for corrosion-resistant alloys; pitting resistance by material choice
GRP	Not applicable (liner thickness per the manufacturer, minimum 2.5 mm resin-rich liner)	Design life 30 years per manufacturer's declaration

9. Consistency with decisions and the equipment register

9.1 The cycle chemistry (AVT(O) with ammonia), the chlorination scheme and the water treatment concept of DEC-EPCE-0015 shall govern the selection of materials in contact with the chemicals; no copper alloys shall be used in the feedwater, condensate and steam systems, in ammonia, or in the wetted parts downstream of ammonia dosing.

9.2 The material entries of the equipment register (equipment.material) have been reviewed against this report. Inconsistencies found are listed as findings of this document and shall be resolved by update of the equipment register or by a technical query. Vendors shall confirm the materials on the datasheets with reference to the clauses of this report.

9.3 Register review result: (1) all seawater-wetted pumps, screens, valves and strainers are listed as super duplex, rubber-lined carbon steel or titanium, consistent with section 4 (strainer elements fixed to super duplex UNS S32750); (2) no equipment in ammonia service is listed with copper alloys; the ammonia storage tanks are fixed to carbon steel; (3) the bronze-fitted fire water pumps and valves (00SGA, 00SGC) are consistent with the fresh-water fire water of DEC-EPCE-0018; (4) the entries without material (00BFT50, 00BFT60, 00SGY10, 10HNE10) have been completed.

9.4 Materials of equipment tagged with a choice (GRP or HDPE for hypochlorite storage; GRP or HDPE-lined carbon steel for chemical tanks) shall be fixed by the vendor proposal and recorded before the datasheet is issued for construction.

10. Inspection, testing and material control

10.1 Positive material identification (PMI) shall be carried out for 100 % of alloy piping, pressure parts of alloy materials and super duplex components; the tests shall be recorded and the result included in the material dossier.

10.2 Super duplex and duplex materials shall meet the ferrite content of 35 to 55 % (30 to 65 % in welds), and shall be tested for intermetallic phases (ASTM A923 method C) and the critical pitting temperature (ASTM G48 method A) with acceptance above 40 degC for super duplex.

10.3 Coating and lining of seawater equipment shall be inspected by holiday testing (spark test) of rubber linings and of coatings on the immersed surfaces before shipment and after installation.

10.4 A corrosion monitoring programme shall be provided for the seawater systems: coupons or probes at the CW intake, at the condenser inlet and at the outfall; wall thickness surveys of the carbon steel piping (steam, feedwater, wet lines) at the first outage and at 5-year intervals, with the flow-accelerated corrosion (FAC) locations identified in the pipe stress and critical line list.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
VGB-S-010	Feed water, boiler water and steam quality for power plants
EN 10204	Metallic products - Types of inspection documents
API 650	Welded Tanks for Oil Storage
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments

